

ON STOCHASTIC PROXIMAL GRADIENT ALGORITHMS

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ABSTRACT. We study a stochastic version of the proximal gradient algorithm where the gradient is analytically intractable, and is approximated by Monte Carlo simulation. We derive a non-asymptotic bound on the convergence rate, and we derive conditions involving the Monte Carlo batch-size and the step-size of the algorithm under which convergence is guaranteed. In particular, we show that the stochastic proximal gradient algorithm achieves the same convergence rate as its deterministic counterpart. We extend the analysis to a stochastic version of the fast iterative shrinkage-thresholding of Beck and Teboulle (2009), whereby the gradient is approximated by Monte Carlo simulation. Contrary to the deterministic setting where the fast iterative shrinkage-thresholding is known to converge faster than the proximal gradient algorithm, our results suggest that in the stochastic setting, the acceleration scheme yields no improvement. To illustrate, we apply the algorithms to the estimation of network structures from multiple realizations of a Gibbs measure.

1. INTRODUCTION

This paper deals with statistical optimization problems of the form:

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$$(\mathbf{P}) \quad \text{Argmin}_{\theta \in \Theta} \{ -\ell(\theta) + g(\theta) \} ,$$

where ℓ is a smooth concave log-likelihood function or some other smooth statistical learning function, and g is a possibly non-smooth convex penalty term.

This problem has attracted a lot of attention with the growing need to address high-dimensional statistical problems (see e.g. the monograph Bühlmann and van de Geer (2011)). This work focuses on the case where the function ℓ and its gradient $\nabla \ell$ are intractable, and where $\nabla \ell$ is given by

$$\nabla \ell(\theta) = \int H(\theta, x) \pi_\theta(dx), \tag{1}$$

for some probability measure π_θ on some measurable space $(\mathbf{X}, \mathcal{B})$, and some measurable function $H : \Theta \times \mathbf{X} \rightarrow \Theta$.

Optimization problems with intractable gradient are well-known in statistics and operation research, and have generated an extensive literature stemming from the seminal work by Robbins and Monro (1951) and Kiefer and Wolfowitz (1952). But these so-called stochastic approximation algorithms cannot handle cases considered in this paper where the function g is non-smooth. To cope with problems where $-\ell + g$ is intractable and possibly non-smooth, various extensions of well-known deterministic algorithms have recently appeared. Some of these work focused on stochastic sub-gradient and mirror descent algorithms; see Nemirovski et al. (2008); Duchi et al. (2011); Lan (2012); Juditsky and Nemirovski (2012a,b). These algorithms operate by constructing stochastic approximations of the sub-gradient of $-\ell + g$. Other authors have proposed algorithms based on proximal operators to better exploit the smoothness of ℓ and the properties of g (Hu et al. (2009); Xiao (2010)). The current paper belongs to this latter thread of work. However the literature cited above is mainly concerned with problems in which samples are generated (either sequentially or in batch) from some unknown distribution and the goal is to minimize some parameter-dependent functional of the distribution. In such scenarios, the gradient takes the form (1), for a sampling

distribution π_θ that does not depend on θ . In statistical problems, intractable likelihood functions often arise because the likelihood is known only up to a normalization factor, depending on the parameter to estimate, which cannot be computed explicitly. This is the case when considering for example likelihood inference for Gibbs measure: in this case, π_θ is the Gibbs measure, known only up to the partition function (normalization constant). Intractable likelihood functions also arise when dealing with hierarchical latent variable models, including missing data or mixed effects models. In that case, the distribution π_θ in (1) represents the conditional distribution of the latent/missing variables given the parameter and the data. In both cases, π_θ not only depends on θ , but is often difficult to simulate and typically requires to use Markov Chain Monte Carlo (MCMC) methods.

In this paper, we study a stochastic version of the proximal gradient algorithm to solve problem (P). Proximal algorithms are well established optimization algorithms for dealing with non-smooth objective functions; Beck and Teboulle (2010); Parikh and Boyd (2013); Juditsky and Nemirovski (2012a,b). In the stochastic version of the proximal gradient algorithm proposed here, the gradient $\nabla\ell(\theta)$ is replaced by an approximation obtained by simulating from π_θ , typically using Markov Chain Monte Carlo (MCMC). We insist again to point out that the optimization problem (P) considered here differs in one crucial aspect from classical stochastic optimization problems. In stochastic optimization problems typically encountered in machine learning and stochastic programming, the distribution π_θ (denoted π as it does not depend on θ) is unknown, and the user is only provided with random samples from π by an oracle. Whereas in the present setting the distribution π_θ is known albeit difficult to simulate, and Monte Carlo simulation (or MCMC simulation) is performed to estimate $\nabla\ell(\theta)$ at a precision controlled (to a large extent) by the user. In this latter setting it is clear that the performance of the stochastic proximal algorithm depends on the possibly varying precision of the gradient estimates. Very little work has been done to incorporate gradient estimation precision in the analysis of stochastic optimization algorithms. One

notable exception is the recent work of Duchi et al. (2012), dealing with composite mirror descent algorithms. Beside the fact that Duchi et al. (2012) deals with the classical framework where the probability measure π_θ does not depend on θ , this work only covers the stochastic mirror descent algorithm (where estimates of the sub-gradient of $-\ell + g$ are employed) which differs from the proximal gradient algorithm used here.

We consider an averaging scheme of the stochastic proximal gradient algorithm and show under some regularity conditions that when the mean squared errors of the gradient estimates decrease like $O(n^{-\delta})$ for some $\delta \geq 1$, the convergence rate of the averaging scheme is $O(n^{-1})$, where n is the number of iterations. This convergence rate is the same as the convergence rate of the deterministic proximal gradient algorithm (Beck and Teboulle (2009)). We extend the analysis to a stochastic version of the fast iterative shrinkage-thresholding algorithm of Beck and Teboulle (2009) and we show that this Nesterov's type accelerated version of the stochastic proximal gradient algorithm achieves the convergence rate of $O(n^{-2})$, when the mean squared errors of the gradient estimates decays like $O(n^{-3-\kappa})$, for any $\kappa > 0$. The optimal convergence rate of the (deterministic) fast iterative shrinkage-thresholding algorithm is $O(n^{-2})$ (Beck and Teboulle (2009)).

However, since these algorithms require the approximation of the gradient at increasing precision (which is costly), the convergence rates depicted above are incomplete as they do not account for the cost of approximating the gradient. A better measure of the performance of these algorithms is the total number of Monte Carlo samples needed to approximate the solution at a given precision $\epsilon > 0$. Our results imply that the number of Monte Carlo samples needed to achieve a precision ϵ is $O(\epsilon^{-2})$ for the stochastic proximal algorithm and $O(\epsilon^{-2-\kappa})$ for some $\kappa > 0$, for the stochastic version of the fast iterative shrinkage-thresholding algorithm. Assuming that these bounds are tight, our analysis suggests that in the stochastic setting and unlike the deterministic setting, the acceleration scheme yields no improvement in general.

We illustrate these results with the problem of estimating the network structure between p nodes using multiple realizations from a Gibbs measure. The simulation results confirm the theoretical findings described above, but also give evidence that the stochastic proximal gradient algorithm is easy to implement and work well in practice.

The rest of the paper is organized as follows. In section 2 our basic assumptions are stated, the proximal gradient algorithm is described and its main properties are recalled. In section 3, the stochastic proximal gradient algorithm is introduced and its performance is studied. In section 4, the accelerated version is presented and analyzed. In particular, the results outlined heuristically above are stated in Theorem 5, Theorem 9, and their corollaries. An application of the stochastic proximal gradient algorithm to the ℓ^1 -penalized inference of a discrete Gibbs measure is presented in section 5. Monte Carlo experiments supporting our findings are reported in Section 5.2. The proofs are postponed to section 6.

2. THE PROXIMAL GRADIENT ALGORITHM

In the sequel Θ denotes a finite-dimensional Euclidean space with norm $\|\cdot\|$ and inner product $\langle \cdot, \cdot \rangle$. Let $\ell : \Theta \rightarrow \mathbb{R}$, and $g : \Theta \rightarrow (-\infty, +\infty]$. We consider the problem of finding solutions to solve (P). In the context of this paper, ℓ represents a log-likelihood, and g a penalty function. We make the following assumption.

H1. *g is convex, not identically $+\infty$, and lower-semicontinuous. The function ℓ is concave and is twice continuously differentiable. In addition, there exists a finite constant L such that, for all $\theta, \theta' \in \Theta$,*

$$\|\nabla \ell(\theta) - \nabla \ell(\theta')\| \leq L \|\theta - \theta'\| ,$$

where $\nabla \ell$ denotes the gradient of ℓ .

H2. *$\min_{\theta \in \Theta} \{-\ell(\theta) + g(\theta)\}$ is finite and attained at $\theta_* \in \Theta$ (not necessarily unique).*

For $\gamma > 0$, the proximal map associated to γg is given by:

$$\text{Prox}_\gamma(\theta) \stackrel{\text{def}}{=} \text{Argmin}_{\vartheta \in \Theta} \left\{ g(\vartheta) + \frac{1}{2\gamma} \|\vartheta - \theta\|^2 \right\}. \quad (2)$$

Parikh and Boyd (2013) gives a nice introduction to proximal operators and algorithms. The proximal gradient algorithm itself can be motivated in a number of ways. Here we follow the Majoration-Maximization (MM) approach described by Beck and Teboulle (2009). Suppose that θ is a working solution of (P). Consider the surrogate function $\vartheta \mapsto Q_\gamma(\vartheta|\theta)$ given by

$$\begin{aligned} Q_\gamma(\vartheta|\theta) &= -\ell(\theta) - \langle \nabla \ell(\theta), \vartheta - \theta \rangle + \frac{1}{2\gamma} \|\vartheta - \theta\|^2 + g(\vartheta) \\ &= -\ell(\theta) + \frac{1}{2\gamma} \|\vartheta - (\theta + \gamma \nabla \ell(\theta))\|^2 - \frac{\gamma}{2} \|\nabla \ell(\theta)\|^2 + g(\vartheta), \end{aligned}$$

for a positive constant $\gamma \in (0, 1/L)$, where L is defined in H1. By construction, $-\ell(\theta) + g(\theta) = Q_\gamma(\theta|\theta)$ for all $\theta \in \Theta$ and $-\ell(\vartheta) + g(\vartheta) \leq Q_\gamma(\vartheta|\theta)$ for all $\theta, \vartheta \in \Theta$. Hence, $\vartheta \mapsto Q_\gamma(\vartheta|\theta)$ is a *majorizing function* for $\theta \mapsto -\ell(\theta) + g(\theta)$ which suggests a MM-type (Majorization-Minorization) recursion, consisting in iteratively computing, the minimum of the majorizing surrogate. This leads to the very popular proximal gradient algorithm, defined below:

Algorithm 1 (Proximal gradient algorithm). Let $\theta_0 \in \Theta$ denote the starting estimate, and $\{\gamma_n, n \in \mathbb{N}\}$, a sequence of positive step sizes. Given θ_n , compute

$$\theta_{n+1} = \text{Prox}_{\gamma_{n+1}}(\theta_n + \gamma_{n+1} \nabla \ell(\theta_n)) .$$

This type of scheme is also known as a forward-backward splitting algorithm: it can be broken up into a forward (explicit) gradient step using the function $\ell(\theta)$, and a backward (implicit) step using the penalization function g ; see (Parikh and Boyd, 2013, Section 4.2). Under H1-H2, $\nabla \ell$ is Lipschitz continuous with constant L and this method can be shown to converge with rate $O(n^{-1})$ where n is the number of iterations, when a fixed stepsize $\gamma \in (0, 1/L]$ is used (as discussed in Combettes and Pesquet (2011)). The method actually converges for stepsizes smaller than $2/L$,

although for stepsizes larger than $1/L$, it can no longer be motivated as MM algorithm, as done above.

When L is not known, the stepsize γ_k can be found by line search; see for example Beck and Teboulle (2010). The algorithm can also be justified from the viewpoint that the solutions of (P) are precisely fixed points of the map $\theta \mapsto \text{Prox}_\gamma(\theta + \gamma \nabla \ell(\theta))$, for γ small enough. This result, as well as existence and uniqueness of solution to problem (P) are summarized in the proposition below (see (Bauschke and Combettes, 2011, Section 27.3) for a proof).

Proposition 1. *Assume H1-H2 and take $\gamma \in (0, 1/L]$.*

- (1) *Every sequence $\{\theta_k, k \in \mathbb{N}\}$ generated by Algorithm 1 applied with the constant sequence $\gamma_k = \gamma$ converges to a solution to problem (P).*
- (2) *Any minimizer $\theta_\star$ of the function $\theta \mapsto -\ell(\theta) + g(\theta)$ satisfies*

$$\theta_\star = \text{Prox}_\gamma(\theta_\star + \gamma \nabla \ell(\theta_\star)) .$$

- (3) *If $-\ell + g$ is strictly convex (which is the case if $-\ell$ or g is strictly convex), then problem (P) admits a unique solution.*

There are many important examples of penalty functions for which the proximal map is tractable.

Example 2. Let $\mathcal{C}$ be a closed convex subset of Θ . Consider the case where $g(\vartheta) = \mathbb{I}_{\mathcal{C}}(\vartheta)$, where

$$\mathbb{I}_{\mathcal{C}}(\vartheta) = \begin{cases} 0 & \text{if } \vartheta \in \mathcal{C} , \\ +\infty & \text{otherwise .} \end{cases} \quad (3)$$

In this case, problem (P) corresponds to finding $\text{Argmin}_{\theta \in \mathcal{C}} \{-\ell(\theta)\}$, and the proximal operator is the orthogonal projection on $\mathcal{C}$, that is the map $\Pi_{\mathcal{C}} : \Theta \rightarrow \Theta$ such that $\|\theta - \Pi_{\mathcal{C}}(\theta)\| = \min_{\vartheta \in \mathcal{C}} \|\theta - \vartheta\|$.

□

Example 3. Suppose $\Theta = \mathbb{R}^p$. A common example of penalty is the elastic-net penalty

$$g(\theta) = \lambda \left(\frac{1-\alpha}{2} \|\theta\|_2^2 + \alpha \|\theta\|_1 \right),$$

for a nonnegative regularization parameter λ ; for $r \geq 1$, $\|\theta\|_r = (\sum_{i=1}^p |\theta_i|^r)^{1/r}$. The parameter $\alpha \in [0, 1]$ controls the balance between the ℓ^1 and ℓ^2 penalties. $\alpha = 1$ corresponds to the lasso penalty, $\alpha = 0$ corresponds to the ridge-regression penalty. For this choice, $\text{Prox}_\gamma(\theta)$ is the component-wise soft-thresholding operator defined as

$$\mathbf{s}_{\gamma, \lambda, \alpha}(\theta)_i = \begin{cases} \frac{\theta_i - \gamma \lambda \alpha}{1 + \gamma \lambda (1 - \alpha)} & \text{if } \theta_i \geq \gamma \lambda \alpha, \\ \frac{\theta_i + \gamma \lambda \alpha}{1 + \gamma \lambda (1 - \alpha)} & \text{if } \theta_i \leq -\gamma \lambda \alpha, \\ 0 & \text{if } \theta_i \in (-\gamma \lambda \alpha, \gamma \lambda \alpha). \end{cases}$$

□

Example 4. Again suppose that $\Theta = \mathbb{R}^p$, and define the rectangle $B \stackrel{\text{def}}{=} \{x = (x_1, \dots, x_p) \in \mathbb{R}^p : a_i \leq x_i \leq b_i, 1 \leq i \leq p\}$, for given $-\infty < a_j < b_j < \infty$. Another interesting choice of penalty is

$$g(\theta) = g_1(\theta) + \mathbb{I}_B(\theta),$$

which combines the hard thresholding $\mathbb{I}$ and a "soft" regularizer g_1 . With this choice, problem (P) becomes

$$\text{Argmin}_{\theta \in \mathbb{R}^p} \{-\ell(\theta) + g_1(\theta) + \mathbb{I}_B(\theta)\} = \text{Argmin}_{\theta \in B} \{-\ell(\theta) + g_1(\theta)\}. \quad (4)$$

This penalty function is also interesting from the computational viewpoint, as it automatically guarantees stability (defined loosely as the propensity of the algorithm to produce a sequence of estimators that remains in a compact set), a key requirement for convergence. When g_1 is the elastic-net penalty of Example 3, the proximal operator associated to g can be computed. Let Π_B denotes the component-wise projection on B . For any $\theta \in \mathbb{R}^p$ and $\gamma > 0$ we have

$$\text{Argmin}_{\vartheta \in \mathbb{R}^p} \left\{ g_1(\vartheta) + \mathbb{I}_B(\vartheta) + \frac{1}{2\gamma} \|\vartheta - \theta\|^2 \right\} = \Pi_B(\mathbf{s}_{\gamma, \lambda, \alpha}(\theta)).$$

3. STOCHASTIC PROXIMAL GRADIENT ALGORITHMS

We consider in this section the setting where $\nabla\ell(\theta)$ is intractable. In most cases, this implies that ℓ is also intractable. We propose a stochastic proximal gradient algorithm whereby the gradient $\nabla\ell(\theta)$ is approximately computed: at iteration n of the algorithm we assume that $\nabla\ell(\theta_n)$ is approximated by $H_{n+1} \in \Theta$. The use of this approximate gradient leads to the following perturbed version of Algorithm 1.

Algorithm 2 (Stochastic Proximal gradient algorithm). Let $\theta_0 \in \Theta$ be the initial solution, $\{\gamma_n\}$ be a sequence of positive non-increasing step sizes, and $\{a_n, n \in \mathbb{N}\}$ be a sequence of nonnegative weights. For $n \geq 1$, given $(\theta_1, \dots, \theta_n)$:

- (1) Obtain $H_{n+1} \in \Theta$, the random approximation of $\nabla\ell(\theta_n)$.
- (2) Compute $\theta_{n+1} = \text{Prox}_{\gamma_{n+1}}(\theta_n + \gamma_{n+1}H_{n+1})$.
- (3) Compute the averaged estimator

$$\bar{\theta}_n \stackrel{\text{def}}{=} \frac{\sum_{k=1}^n a_k \theta_k}{\sum_{k=1}^n a_k} = \left(1 - \frac{a_n}{\sum_{k=1}^n a_k}\right) \bar{\theta}_{n-1} + \frac{a_n}{\sum_{k=1}^n a_k} \theta_n . \quad (5)$$

To describe the behavior of Algorithm 2 , we introduce some additional notations. Set $\mathcal{F}_n \stackrel{\text{def}}{=} \sigma(\theta_0, H_1, \dots, H_n)$. The sequence $\{\theta_n, n \in \mathbb{N}\}$ specified by Algorithm 2 can be rewritten as

$$\theta_{n+1} = \text{Prox}_{\gamma_{n+1}}(\theta_n + \gamma_{n+1} \nabla\ell(\theta_n) + \gamma_{n+1} \eta_{n+1}) ,$$

where η_{n+1} is the *approximation error*:

$$\eta_{n+1} \stackrel{\text{def}}{=} H_{n+1} - \nabla\ell(\theta_n) . \quad (6)$$

Two quantities will appear naturally when studying the convergence of this algorithm:

$$\epsilon_n^{(1)} \stackrel{\text{def}}{=} \|\mathbb{E}[\eta_{n+1} | \mathcal{F}_n]\| , \quad \text{and} \quad \epsilon_n^{(2)} \stackrel{\text{def}}{=} \mathbb{E} \left[\|\eta_{n+1}\|^2 \mid \mathcal{F}_n \right] . \quad (7)$$

The following theorem provides a control of the mean value $\mathbb{E}[V(\bar{\theta}_n)]$ where the function V is defined by

$$V(\theta) \stackrel{\text{def}}{=} -\ell(\theta) + g(\theta) - (-\ell(\theta_\star) + g(\theta_\star)) . \quad (8)$$

Theorem 5. *Assume H1-H2. Suppose that $0 < L\gamma_n \leq 1$ for all $n \geq 1$. Let $\{\bar{\theta}_n, n \geq 0\}$ be the averaged estimator given by (5) for some sequence $\{a_n, n \geq 0\}$ of nonnegative numbers such that $a_0 = 0$. Then for all $n \geq 1$,*

$$\begin{aligned} \left(\sum_{j=1}^n a_j \right) \mathbb{E} [V(\bar{\theta}_n)] &\leq \frac{1}{2} \sum_{j=1}^n \left(\frac{a_j}{\gamma_j} - \frac{a_{j-1}}{\gamma_{j-1}} \right) \mathbb{E} [\|\theta_{j-1} - \theta_\star\|^2] \\ &\quad + \sum_{j=1}^n a_j \mathbb{E} [\|\theta_{j-1} - \theta_\star\| \epsilon_{j-1}^{(1)}] + \sum_{j=1}^n a_j \gamma_j \mathbb{E} [\epsilon_{j-1}^{(2)}] . \end{aligned} \quad (9)$$

Proof. See Section 6.1. □

We now compare the rate of convergence of the averaged sequence $\{\bar{\theta}_n, n \in \mathbb{N}\}$ obtained from Algorithm 2 to the rate of convergence of the deterministic proximal gradient algorithm (Beck and Teboulle (2009)), in the situation when $\nabla \ell(\theta)$ is defined as the expectation of a function:

H3. *For all $\theta \in \Theta$,*

$$\nabla \ell(\theta) = \int_{\mathbf{X}} H(\theta, x) \pi_\theta(dx) , \quad (10)$$

for some probability measure π_θ on a measurable space $(\mathbf{X}, \mathcal{B})$, and a measurable function $H : \Theta \times \mathbf{X} \rightarrow \Theta$, $(\theta, x) \mapsto H(\theta, x)$, satisfying $\int \pi_\theta(dx) \|H(\theta, x)\| < \infty$.

When $\nabla \ell(\theta)$ takes this integral form the optimization problem shares some similarities with the one considered in Duchi et al. (2012), In our setting however the distribution π_θ depends upon the parameter $\theta \in \Theta$. This introduces an additional twist into the problem, that requires to be tackled. In this setting, the approximate gradient H_{n+1} can be computed in a number of different ways. When the dimension of $\mathbf{X}$ is low, numerical integration methods can be used to form H_{n+1} deterministically. When the dimension of $\mathbf{X}$ becomes larger, Monte Carlo integration method can

be used instead ¹; in this case, H_{n+1} is a random variable, which is the main case of interest in this work. For example, if it is possible to sample directly the target distribution π_θ one may set

$$H_{n+1} = m_{n+1}^{-1} \sum_{j=1}^{m_{n+1}} H(\theta_n, X_{n+1,j}) , \quad (11)$$

for some positive sequence of integers $\{m_n, n \in \mathbb{N}\}$, where $\{X_{n+1,j}\}_{j=1}^{m_{n+1}}$ is, conditionally to the past $\mathcal{F}_n$, a sequence of i.i.d. random variables distributed according to the current target distribution π_{θ_n} . We refer to $\{X_{n+1,j}\}_{j=1}^{m_{n+1}}$ as a Monte Carlo batch and m_n is the Monte Carlo batch-size. In this case, the approximation bias $\epsilon_n^{(1)} = 0$ (see (7)), whereas the approximation variance is

$$\epsilon_n^{(2)} = m_{n+1}^{-1} \int \|H(\theta_n, x) - \nabla \ell(\theta_n)\|^2 \pi_{\theta_n}(dx) . \quad (12)$$

As mentioned above, contrary to Duchi et al. (2012), the sampling distribution π_θ depends on θ . It is conceptually possible to transform this problem by importance sampling to the case where the sampling distribution is free of the parameter, i.e. assuming that the family of distributions $(\pi_\theta, \theta \in \Theta)$ is dominated by a common distribution π_* (referred to as the instrumental distribution), to rewrite (10) as

$$\nabla \ell(\theta) = \int_{\mathbf{X}} H(\theta, x) w(\theta, x) \pi_*(dx) , \quad (13)$$

where $w(\theta, \cdot) = d\pi_\theta/d\pi_*$ is the importance weight, i.e. Radon-Nikodym derivative of π_θ with respect to the dominating distribution π_* ; see Geyer and Thompson (1992), Geyer (1994). The importance sampling estimator H_{n+1} of $\nabla \ell(\theta)$ is given in this context by

$$H_{n+1} = m_{n+1}^{-1} \sum_{j=1}^{m_{n+1}} H(\theta_n, X_{n+1,j}) w(\theta_n, X_{n+1,j}) , \quad (14)$$

where $\{X_{n+1,j}\}_{j=1}^{m_{n+1}}$ is, conditionally to the past $\mathcal{F}_n$, a sequence of i.i.d. random variables distributed according to the instrumental distribution π_* . Whereas the collection of function $\{\theta \mapsto H(\theta, x), x \in \mathbf{X}\}$ is typically convex, this is not necessarily the case of the product $\theta \mapsto H(\theta, x) w(\theta, x)$, which is a

¹Quasi Monte Carlo methods are an option when the dimension of $\mathbf{X}$ is intermediate; we do not consider these approximations here, but our analysis can be extended to this setting.

key assumption in Duchi et al. (2012). In this work, on the contrary, we do not make assumptions on the dependence $\theta \mapsto \pi_\theta$; we stress in particular that we do not require any kind of convexity condition for the mapping $\theta \mapsto H(\theta, x)w(\theta, x)$ for all x .

In addition, it is well known from the analysis of importance sampling algorithms that this strategy breaks down when the instrumental distribution π_* is not well fitted to the target π_θ , a situation which typically implies large fluctuation of the importance weights $\{w(\theta_n, X_{n+1,j})\}_{j=1}^{m_{n+1}}$; this occurs all the more as the dimension of the integration space increases. It is possible to alleviate this problem by modifying the instrumental distribution as the parameter θ_n evolves; this is not covered in Duchi et al. (2012).

A more flexible approach is to compute H_{n+1} using Markov chain Monte Carlo (MCMC) algorithms. Let P_θ be a Markov kernel with invariant distribution π_θ , and let ν_θ be some initial distribution on $\mathbf{X}$; see e.g. Robert and Casella (2005) and the references therein. At the n -th iteration, we compute H_{n+1} as the empirical mean

$$H_{n+1} = m_{n+1}^{-1} \sum_{j=1}^{m_{n+1}} H(\theta_n, X_{n+1,j}) ,$$

where $\{X_{n+1,j}\}_{j=1}^{m_{n+1}}$ is, conditionally to the past $\mathcal{F}_n$, a Markov chain with initial distribution ν_{θ_n} and Markov kernel P_{θ_n} . In many instances the kernel is geometrically ergodic with ergodicity constants which are controlled uniformly over $\theta \in \mathcal{K}$ where $\mathcal{K}$ is a compact subset of Θ , i.e. $\sup_{\theta \in \mathcal{K}} \|\nu_\theta P_\theta^j H(\theta, \cdot) - \pi_\theta H(\theta, \cdot)\| \leq C_{\mathcal{K}} \rho_{\mathcal{K}}^j$ with $\rho_{\mathcal{K}} \in [0, 1)$ and $C_{\mathcal{K}} < \infty$; see for example (Fort et al., 2011, Lemma 2.3.).

In this setting, the bias $\epsilon_n^{(1)}$ can be computed as

$$\mathbb{E}[\eta_{n+1} | \mathcal{F}_n] = m_{n+1}^{-1} \sum_{j=0}^{m_{n+1}-1} \left(\nu_{\theta_n} P_{\theta_n}^j H(\theta_n, \cdot) - \pi_{\theta_n} H(\theta_n, \cdot) \right) ,$$

so that

$$\epsilon_n^{(1)} \mathbb{1}_{\mathcal{K}}(\theta_n) \leq \left(\sum_{j=0}^{m_{n+1}-1} \sup_{\theta \in \mathcal{K}} \|\nu_\theta P_\theta^j H(\theta, \cdot) - \pi_\theta H(\theta, \cdot)\| \right) m_{n+1}^{-1} . \quad (15)$$

Furthermore, by (Fort and Moulines, 2003, Proposition 12)

$$\epsilon_n^{(2)} \mathbb{1}_{\mathcal{K}}(\theta_n) \leq \left(6(1 + C_r) \sup_{\theta \in \mathcal{K}} \sup_{j \geq 0} \nu_{\theta} P_{\theta}^j G_{\theta} \right) m_{n+1}^{-1},$$

where $G_{\theta}(x) \stackrel{\text{def}}{=} \|\sum_{k \geq 0} P_{\theta}^k (H(\theta, \cdot) - \pi_{\theta} H(\theta, \cdot))(x)\|^2$ and C_r is the Rosenthal constant (see e.g. (Hall and Heyde, 1980, Theorem 2.12)). Sufficient conditions for these upper bounds to be finite can be found in (Fort et al., 2011, Lemma 2.3.).

The next corollary provides appropriate conditions on the approximation error η_{n+1} so that the averaging scheme $\bar{\theta}_n$ given by Algorithm 2 achieves the same convergence rate $1/n$ as the deterministic proximal gradient algorithm (Beck and Teboulle (2009)). The proof follows easily from Theorem 5.

Corollary 6. *In addition to the assumptions of Theorem 5, suppose H3 and there exist constants $C_1, C_2, B < \infty$ such that, for $n \geq 1$,*

$$\mathbb{E}[\epsilon_n^{(1)}] \leq C_1 m_{n+1}^{-1}, \quad \mathbb{E}[\epsilon_n^{(2)}] \leq C_2 m_{n+1}^{-1}, \quad \text{and} \quad \sup_{n \in \mathbb{N}} \|\theta_n - \theta_{\star}\| \leq B, \quad \mathbb{P} - a.s. \quad (16)$$

Then, for $n \geq 1$,

$$\mathbb{E}[V(\bar{\theta}_n)] \leq \left(\sum_{j=1}^n a_j \right)^{-1} \left\{ \frac{B^2 a_n}{2\gamma_n} + BC_1 \sum_{j=1}^n a_j m_j^{-1} + C_2 \sum_{j=1}^n a_j \gamma_j m_j^{-1} \right\}.$$

Let us discuss the rate when

$$a_n = C_{\alpha} n^{\alpha} \quad m_n = C_{\beta} n^{\beta} \quad \gamma_n = C_{\gamma} n^{-\kappa},$$

for some $\alpha, \beta, \kappa \geq 0$. This discussion will cover the case of weighted (resp. uniform) averaging in $\bar{\theta}_n$ by choosing $\alpha > 0$ (resp. $\alpha = 0$); the case of constant (resp. increasing) Monte Carlo batch-size by choosing $\beta = 0$ (resp. $\beta > 0$); and the case of constant (resp. vanishing) stepsize sequence by

choosing $\kappa = 0$ (resp. $\kappa > 0$). By Corollary 6 we have

$$(\alpha + 1)^{-1} \mathbb{E} [V(\bar{\theta}_n)] \leq \frac{B^2}{2C_\gamma} \frac{1}{n^{1-\kappa}} + \frac{BC_1}{C_\beta} \frac{1}{n^{\alpha+1}} \sum_{j=1}^n \frac{1}{j^{\beta-\alpha}} + \frac{C_2C_\gamma}{C_\beta} \frac{1}{n^{\alpha+1}} \sum_{j=1}^n \frac{1}{j^{\beta-\alpha+\kappa}}. \quad (17)$$

The behavior of $\bar{\theta}_n$ depends on whether the gradient approximation H_n is unbiased or not.

3.1. Convergence rate of $\bar{\theta}_n$: unbiased gradient approximation. Assume that the approximation bias of $\nabla \ell(\theta_n)$ is null (i.e. $C_1 = 0$) and the number of simulations is slowly increasing, i.e., $\beta \in [0, \alpha + 1 - \kappa)$. In this setting, $\mathbb{E} [V(\bar{\theta}_n)] \leq U_n$ where as n goes to infinity,

$$U_n \sim \frac{(\alpha + 1)B^2}{2C_\gamma n^{1-\kappa}} + \frac{(\alpha + 1)}{(\alpha + 1 - \beta - \kappa)} \frac{C_2C_\gamma}{C_\beta} \frac{1}{n^{\beta+\kappa}}. \quad (18)$$

By choosing $\kappa = (1 - \beta)/2$ and either $\alpha = 0$ and $\beta \in [0, 1)$ or $\alpha > 0$ and $\beta \in [0, 1]$, this yields

$U_n \sim D/n^{(\beta+1)/2}$ where

$$D = (\alpha + 1) \left(\frac{B^2}{2C_\gamma} + \frac{C_2C_\gamma}{C_\beta(\alpha + 1 - (\beta + 1)/2)} \right).$$

Assume that the desired accuracy for the estimator is $\delta > 0$, i.e. we would like to guarantee that $\mathbb{E} [V(\bar{\theta}_n)] \leq \delta$. Then we need $n \geq (D\delta^{-1})^{2/(\beta+1)}$ and the total number of Monte Carlo samples required up to time n , which is $\sum_{j=1}^n m_j \sim C_\beta(\beta + 1)^{-1}n^{\beta+1}$, increases like $C_\beta(\beta + 1)^{-1}(D\delta^{-1})^2$.

This discussion holds even when $\beta = 0$, thus showing that when the gradient approximation is unbiased, the number of simulations at each iteration can be chosen constant ($\beta = 0$) and both the uniformly averaged and the weighted averaged (resp. $\alpha = 0$ and $\alpha > 0$) stochastic proximal gradient algorithms achieve the convergence rate of $n^{-1/2}$ by choosing a decreasing step-size sequence $\gamma_n = C_\gamma/\sqrt{n}$. The weight sequence a_n does not play a role in the rate of convergence: any non-negative value of α is convenient; for example, α can be fixed to the value which minimizes the constant D .

3.2. Convergence rate of $\bar{\theta}_n$: biased gradient approximation. Let us now discuss the rate of convergence when the approximation bias is non-zero (i.e. $C_1 > 0$). In the case $C_1 > 0$, we see from Equation 17 that the number of samples m_n per iterations can not be constant and has to increase as a function of n (i.e. $\beta > 0$). We restrict the discussion to the case the number of simulations is slowly increasing, i.e., $\beta \in (0, \alpha + 1 - \kappa)$. In this setting, $\mathbb{E}[V(\bar{\theta}_n)] \leq U_n$ where as n goes to infinity,

$$U_n \sim \frac{D_1}{n^{1-\kappa}} + \frac{D_2}{n^\beta} + \frac{D_3}{n^{\beta+\kappa}} \quad (19)$$

where

$$D_1 \stackrel{\text{def}}{=} \frac{(\alpha + 1)B^2}{2C_\gamma}, \quad D_2 \stackrel{\text{def}}{=} \frac{(\alpha + 1)BC_1}{(\alpha + 1 - \beta)C_\beta}, \quad D_3 \stackrel{\text{def}}{=} \frac{(\alpha + 1)C_2C_\gamma}{(\alpha + 1 - \beta - \kappa)C_\beta}.$$

Assume first that the stepsize is constant $\gamma_n = C_\gamma$ (i.e. $\kappa = 0$). Then, $\mathbb{E}[V(\bar{\theta}_n)] \leq U_n$ where as n goes to infinity, $U_n \sim D_1/n + D/n^\beta$ with $D = D_2 + D_3$. When $\beta \in (0, 1)$, $U_n \sim D/n^\beta$ as $n \rightarrow \infty$ for any $\alpha \geq 0$ and for a given accuracy $\delta > 0$, the number of iterations n should be increased such that $n \geq (D\delta^{-1})^{1/\beta}$. The total number of simulations, $\sum_{j=1}^n m_j \sim C_\beta(\beta + 1)^{-1}n^{\beta+1}$ is therefore asymptotically equivalent, as δ goes to zero, to $C_\beta(\beta + 1)^{-1}(D\delta^{-1})^{(\beta+1)/\beta}$. Since $\beta < 1$, this rate is larger than δ^{-2} . When $\beta = 1$ (which implies $\alpha > 0$), $U_n \sim (D_1 + D)/n$ as $n \rightarrow \infty$ and the total number of simulations required to reach the accuracy δ is $C_\beta(D_1 + D)^2\delta^{-2}/2$. Finally, when $\beta \in (1, 1 + \alpha)$ (which implies again $\alpha > 0$) then $U_n \sim D_1/n$ as $n \rightarrow \infty$ and the total number of simulations required to reach the accuracy δ is proportional to $\delta^{-(\beta+1)}$. This discussion shows that when the step size is kept fixed, it is optimal to consider the weighted averaged stochastic proximal gradient algorithm and to increase linearly the Monte Carlo batch size m_n with the iterations; in this case, the computation cost to reach the accuracy δ is proportional to δ^{-2} and the averaged estimator converges at the rate $1/n$.

Let us now discuss the vanishing stepsize case, $\gamma_n = C_\gamma n^{-\kappa}$ with $\kappa \in (0, 1)$. First, observe that in this case, $\mathbb{E}[V(\bar{\theta}_n)]$ goes to zero with a rate at most $1/n^{1-\kappa}$ which is slower than $1/n$. When $\beta \in (0, 1 - \kappa]$, U_n vanishes at the rate $1/n^\beta$ and the total number of simulations required

to reach the accuracy δ is proportional to $(\delta^{-1})^{(\beta+1)/\beta}$. This number is minimal when $\beta = 1 - \kappa$. When $\beta \in (1 - \kappa, 1 + \alpha - \kappa)$ (which implies $\alpha > 0$), $U_n \sim D_1/n^{1-\kappa}$ and the total number of simulations required to reach the accuracy δ is given by $C_\beta(\beta+1)^{-1}(D_1\delta^{-1})^{(\beta+1)/(1-\kappa)}$. Note that since $\beta > 1 - \kappa > 0$, $(\beta+1)/(1-\kappa) > (\beta+1)/\beta$. This discussion shows that when the step size is vanishing, it is optimal to increase the Monte Carlo batch size m_n as $n\gamma_n$; in this case, the computational cost for both the weighted averaged algorithm and the uniform averaged is proportional to $\delta^{-(2-\kappa)/(1-\kappa)} = \delta^{-(1+\beta)/\beta}$ and the rate of convergence is proportional to $1/n^{1-\kappa} = 1/n^\beta$; whatever $\kappa \in (0, 1)$.

Therefore, since $2 < (\beta+1)/\beta$ when $\beta < 1$, the above discussions show that when the bias is non zero, a vanishing step size sequence yields a slower rate of convergence and increases the computational cost.

3.3. Convergence of the non-averaged estimate. The next theorem shows that the sequence $\{\theta_n, n \geq 0\}$ itself also converges under some very mild assumptions. We impose the following condition.

H4. *There exists $p \geq 2$ such that for all compact subset $\mathcal{K}$ of Θ ,*

$$\sum_{j \geq 1} \mathbb{E}[\|\eta_{j+1}\|^p \mathbb{1}_{\mathcal{K}}(\theta_j)] < \infty.$$

Remark 7. In Monte Carlo simulation or MCMC simulation with geometrically ergodic Markov kernel it is typically the case that $\mathbb{E}[\|\eta_{j+1}\|^p \mathbb{1}_{\mathcal{K}}(\theta_j)] \leq C m_{j+1}^{-p/2}$, for some finite constant C (see e.g. (Fort and Moulines, 2003, Proposition 12)). Hence H4 is a fairly mild requirement which implies $m_j \uparrow \infty$.

Theorem 8. *Assume H1 to H4. Assume also that $-\ell + g$ is continuous. Let $\{\theta_n, n \geq 0\}$ be the sequence given by Algorithm 2 with $\gamma_n = \gamma$ for $\gamma \in (0, 1/L]$. Suppose also $\mathcal{L}_\gamma \stackrel{\text{def}}{=} \{\theta \in \Theta : \theta = \text{Prox}_\gamma(\theta + \gamma \nabla \ell(\theta))\}$ is finite and that $\limsup_n \|\theta_n\|$ is finite almost-surely. Then the sequence $\{\theta_n, n \in \mathbb{N}\}$ converges to an element of $\mathcal{L}_\gamma$ almost surely, as $n \rightarrow \infty$.*

Proof. See Section 6.2. □

4. NESTEROV'S ACCELERATION

Nesterov (1983) introduced an acceleration scheme of the deterministic gradient method that is shown to converge at a rate $O(1/n^2)$. The algorithm was recently extended by Beck and Teboulle (2009) to solve problems of the form (P), where $\nabla\ell$ is tractable. We consider the stochastic version of Beck-Teboulle algorithm to handle problems where $\nabla\ell$ is intractable.

Let $\{t_n, n \in \mathbb{N}\}$ be a sequence of positive numbers with the following property

$$t_0 = 1, \quad t_{n+1}^2(1 - t_{n+1}^{-1}) \leq t_n^2, \quad n \geq 0. \quad (20)$$

For instance (20) holds for all sequences $t_n \propto (n + n_0)^\beta$, with $\beta \in (0, 1)$, for some $n_0 > 0$. A commonly used sequence that satisfies (20) is defined recursively as

$$t_{n+1} = \frac{1 + \sqrt{1 + 4t_n^2}}{2}. \quad (21)$$

It is easily seen that with (21) $t_n \sim (1/2)n$ as n goes to infinity.

Algorithm 3 (Fast Stochastic Proximal gradient algorithm). Let $\theta_0 \in \Theta$, $\{\gamma_n\}$ be a sequence of positive non-increasing step sizes and $\{t_n\}$ be a sequence satisfying (20). Compute $\theta_1 = \text{Prox}_{\gamma_1}(\theta_0 + \gamma_1 H_1)$, where H_1 is an approximation of $\nabla\ell(\theta_0)$. For $n \geq 1$, given $(\theta_0, \dots, \theta_n)$:

(1) Compute

$$\vartheta_n = \theta_n + t_n^{-1}(t_{n-1} - 1)(\theta_n - \theta_{n-1}). \quad (22)$$

(2) Obtain H_{n+1} a random approximation of $\nabla\ell(\vartheta_n)$, and set

$$\theta_{n+1} = \text{Prox}_{\gamma_{n+1}}(\vartheta_n + \gamma_{n+1} H_{n+1}). \quad (23)$$

For this algorithm, the approximation error η_{n+1} is defined by $\eta_{n+1} \stackrel{\text{def}}{=} H_{n+1} - \nabla \ell(\vartheta_n)$. Set $\mathcal{F}_n \stackrel{\text{def}}{=} \sigma(\theta_0, H_1, \dots, H_n)$ and

$$\epsilon_n^{(1)} \stackrel{\text{def}}{=} \|\mathbb{E}[\eta_{n+1} | \mathcal{F}_n]\|, \quad \text{and} \quad \epsilon_n^{(2)} \stackrel{\text{def}}{=} \mathbb{E}[\|\eta_{n+1}\|^2 | \mathcal{F}_n]. \quad (24)$$

Theorem 9. *Assume H1 and H2. Let $\{t_n, n \in \mathbb{N}\}$ be a non-decreasing sequence satisfying (20), and $\{\gamma_n, n \in \mathbb{N}\}$ be a sequence of positive non-increasing step sizes such that $\gamma_0 \in (0, 1/L]$. Let $\{\theta_n, n \geq 0\}$ be the sequence given by Algorithm 3. For all $n \geq 1$, it holds*

$$\begin{aligned} \gamma_{n+1} t_n^2 \mathbb{E}[V(\theta_{n+1})] &\leq \mathbb{E}\left[\gamma_1 V(\theta_1) + \frac{1}{2} \|\theta_1 - \theta_\star\|^2\right] + \sum_{j=1}^n t_j^2 \gamma_{j+1} \mathbb{E}\left[\|\theta_j - \theta_\star\| \epsilon_j^{(1)}\right] \\ &\quad + \sum_{j=1}^n t_j^2 \gamma_{j+1} \mathbb{E}\left[\|\vartheta_j - \theta_\star\| \epsilon_j^{(1)}\right] + \sum_{j=1}^n t_j^2 \gamma_{j+1}^2 \mathbb{E}\left[\epsilon_j^{(2)}\right], \end{aligned} \quad (25)$$

where $\epsilon_j^{(i)}$ are given by (24).

Proof. See Section 6.3. □

Upon noting that, under the assumptions of Theorem 9

$$\|\vartheta_j - \theta_\star\| \leq \|\theta_j - \theta_\star\| + \frac{t_{j-1}}{t_j} (\|\theta_j - \theta_\star\| + \|\theta_{j-1} - \theta_\star\|) \leq 3 \sup_j \|\theta_j - \theta_\star\|,$$

the following corollary is a trivial consequence of Theorem 9.

Corollary 10. *In addition to the assumptions of Theorem 9, suppose that there exist constants $C_1, C_2, B < \infty$ such that, for $n \geq 1$,*

$$\mathbb{E}[\epsilon_n^{(1)}] \leq C_1 m_{n+1}^{-1}, \quad \mathbb{E}[\epsilon_n^{(2)}] \leq C_2 m_{n+1}^{-1}, \quad \text{and} \quad \sup_{n \in \mathbb{N}} \|\theta_n - \theta_\star\| \leq B, \quad \mathbb{P} - a.s. \quad (26)$$

Then, for $n \geq 1$,

$$\mathbb{E}[V(\theta_{n+1})] \leq \frac{1}{\gamma_{n+1} t_n^2} \left(\gamma_1 \mathbb{E}[V(\theta_1)] + \frac{B^2}{2} \right) + \frac{4BC_1}{\gamma_{n+1} t_n^2} \sum_{k=1}^n \frac{\gamma_{k+1} t_k^2}{m_{k+1}} + \frac{C_2}{\gamma_{n+1} t_n^2} \sum_{k=1}^n \frac{\gamma_{k+1}^2 t_k^2}{m_{k+1}}.$$

Let us discuss the rate of convergence when $\gamma_n \sim C_\gamma n^{-\kappa}$ for some $\kappa \geq 0$, and the sequence $\{t_n, n \in \mathbb{N}\}$ is given by (21) so that $t_n \sim n/2$ when $n \rightarrow \infty$.

4.1. Convergence rate of θ_n : unbiased gradient approximation. Assume first the bias in approximating the gradient is null (i.e. $C_1 = 0$). Then $\mathbb{E}[V(\theta_n)] \leq U_n$ and as $n \rightarrow \infty$,

$$U_n \sim \frac{D_1}{n^{2-\kappa}} + \frac{D_2}{n^{2-\kappa}} \sum_{k=1}^n \frac{k^{2-2\kappa}}{m_{k+1}}$$

where $D_1 \stackrel{\text{def}}{=} 4(\mathbb{E}[V(\theta_1)] + B^2/(2C_\gamma))$ and $D_2 \stackrel{\text{def}}{=} C_2 C_\gamma$.

When the step size is kept fixed $\gamma_n = C_\gamma$ (i.e. $\kappa = 0$), the accelerated stochastic algorithm achieves the same rate of convergence of $1/n^2$ as the deterministic accelerated proximal gradient algorithm (Beck and Teboulle (2009)) as soon as $\sum_{k \geq 1} k^2/m_{k+1} < \infty$. This holds true when $m_n \sim C_\beta n^3 \log^\beta n$ for some $\beta > 1$. However, the rate should be related to the computational cost: the number of iterations to reach the accuracy δ is proportional to $\delta^{-1/2}$ and when $m_n \sim C_\beta n^3 \log^\beta n$, the total number of simulations up to iteration n is $\sum_{k=1}^n m_{k+1} \sim n^4 (\log n)^\beta / 4$. Hence, the computational cost is proportional to $\delta^{-2} (-\log \delta)^\beta$, which is larger than the computational cost of the basic algorithm (see Section 3, the rate is proportional to δ^{-2}).

When the step size is vanishing $\gamma_n \sim C_\gamma n^{-\kappa}$ for some $\kappa \in (0, 2)$, the rate of the accelerated stochastic algorithm is at most $1/n^{2-\kappa}$; it is therefore slower than the deterministic one. In addition, when the number of simulations is $m_n \sim C_\beta n^{3-2\kappa} (\log n)^\beta$ the computational cost to reach the accuracy δ is proportional to $\delta^{-2} (-\log \delta)^\beta$. Hence, the computational cost of the accelerated algorithm is the same whatever the strategy for the step-size sequence ($\kappa = 0$ or $\kappa \in (0, 2)$).

As a conclusion, this discussion evidences that when the bias is null, the computational cost to reach the accuracy δ is in favor of the basic stochastic proximal gradient algorithm (compare the rate δ^{-2} for the basic algorithm to $\delta^{-2} (-\log \delta)^\beta$ for the accelerated one).

4.2. Convergence rate of θ_n : biased gradient approximation. Assume now that the bias in approximating the gradient is not null, i.e. $C_1 > 0$. Then $\mathbb{E}[V(\theta_n)] \leq U_n$ and as $n \rightarrow \infty$,

$$U_n \sim \frac{D_1}{n^{2-\kappa}} + \frac{D_2}{n^{2-\kappa}} \sum_{k=1}^n \frac{k^{2-2\kappa}}{m_{k+1}} + \frac{4BC_1}{n^{2-\kappa}} \sum_{k=1}^n \frac{k^{2-\kappa}}{m_{k+1}}.$$

When the step size is kept fixed (i.e. $\kappa = 0$), the discussion is the same as when the bias is null: the rate of convergence of the accelerated algorithm is $1/n^2$ provided the number of simulations increases in such a way that $\sum_{k \geq 1} k^2/m_k < \infty$; and when $m_n \sim C_\beta n^3 (\log n)^\beta$ for some $\beta > 1$, the computational cost to reach the accuracy δ is proportional to $\delta^{-2}(-\log \delta)^\beta$.

When the step size vanishes (i.e. $\kappa \in (0, 2)$), the rate of U_n is $1/n^{2-\kappa}$ provided $\sum_{k \geq 1} k^{2-\kappa}/m_k < \infty$. In the case $m_n \sim C_\beta n^{3-\kappa} (\log n)^\beta$ for some $\beta > 1$, the computational cost to reach the accuracy δ is proportional to $\delta^{-(4-\kappa)/(2-\kappa)}(-\log \delta)^\beta$. Note that since $\kappa > 0$, this computational cost is always larger than $\delta^{-2}(-\log \delta)^\beta$ - which is the computational cost with constant step size; and than δ^{-2} - which is the computational cost of the basic stochastic proximal gradient algorithm with constant step size (see Section 3).

As a conclusion, this discussion evidences that when the bias is not zero, the computational cost to reach the accuracy δ is again in favor of the basic stochastic proximal gradient algorithm (compare the rate δ^{-2} for the basic algorithm to $\delta^{-2}(-\log \delta)^\beta$ for the accelerated one with constant step size).

5. EXAMPLE: HIGH-DIMENSIONAL NETWORK STRUCTURE ESTIMATION

The problem of estimating sparse network structures from measurements on the nodes of the network has generated much research activities in recent years. For continuous measurements this problem is typically formulated as a covariance matrix estimation problem (Drton and Perlman (2004); d'Aspremont et al. (2008); Bickel and Levina (2008)). We focus here on the case where the measurements are discrete, in which case the estimation of the network structure can be framed as estimating the parameters of a Gibbs measure with joint probability mass function

$$f_\theta(x_1, \dots, x_p) = \frac{1}{Z_\theta} \exp \left\{ \sum_{i=1}^p \theta_{ii} B_0(x_i) + \sum_{1 \leq j < i \leq p} \theta_{ij} B(x_i, x_j) \right\}, \quad (27)$$

for a function $B_0 : \mathsf{X} \rightarrow \mathbb{R}$, and a symmetric function $B : \mathsf{X} \times \mathsf{X} \rightarrow \mathbb{R}$, where X is a finite set. The real-valued symmetric matrix θ defines the network structure and is the parameter of interest. It

has the same interpretation as the precision matrix in a multivariate Gaussian distribution. The term Z_θ is the normalizing constant, which cannot (in general) be computed explicitly. Although Z_θ involves a summation over the discrete set $\mathbf{X}^p$, we will write it for convenience as an integral with respect to the counting measure μ on $\mathbf{X}^p$,

$$Z_\theta = \int_{\mathbf{X}^p} \exp \left\{ \sum_{1 \leq j \leq i \leq p} \theta_{ij} B_{ij}(x_i, x_j) \right\} \mu(\mathrm{d}x),$$

where $B_{ij}(x, y) = B(x, y)$ if $i \neq j$, and $B_{ij}(x, y) = B_0(x)$ if $i = j$. We consider the problem of estimating θ from N realizations $\{x^{(i)}\}_{i=1}^N$ from (27) where $x^{(i)} = (x_1^{(i)}, \dots, x_p^{(i)}) \in \mathbf{X}^p$, and where the true value of θ is assumed sparse. This problem is relevant for instance in biology where this model has recently been proposed for estimating molecular co-evolution from multiple aligned sequences (Ekeberg et al. (2013); Kamisetty et al. (2013)). In the statistical and machine learning literature this problem has been considered by many authors (Banerjee et al. (2008); Höfling and Tibshirani (2009); Ravikumar et al. (2010); Guo et al. (2010); Xue et al. (2012)). However so far, the focus has mainly been on the Ising model where $\mathbf{X} = \{0, 1\}$ (or $\{-1, 1\}$) and $B(x, y) = xy$, whereas in the protein co-evolution estimation problem $\mathbf{X} = \{1, \dots, M\}$, where $M = 20$ (sometimes 21) is the number of amino-acids, and $B(x, y) = \mathbb{1}_{\{x=y\}}$. Furthermore, to the exception of Höfling and Tibshirani (2009), most of these work estimate θ by maximizing a ℓ^1 -penalized pseudo-likelihood function, an approach that helps avoid the intractable normalizing constant Z_θ but is known to be sub-optimal. We will see that the stochastic proximal gradient algorithm gives a simple and natural algorithm for computing the ℓ^1 -penalized likelihood estimate of θ , for an arbitrary interaction function B .

Let Θ denote the space of $p \times p$ symmetric matrices equipped with the (modified) Frobenius inner product

$$\langle \theta, \vartheta \rangle \stackrel{\text{def}}{=} \sum_{1 \leq k \leq j \leq p} \theta_{jk} \vartheta_{jk}, \text{ with norm } \|\theta\|_2 \stackrel{\text{def}}{=} \sqrt{\langle \theta, \theta \rangle}.$$

For $x \in \mathbb{X}^p$, we define the matrix $\bar{B}(x) \stackrel{\text{def}}{=} (B_{jk}(x_j, x_k))_{1 \leq j, k \leq p} \in \mathbb{R}^{p \times p}$. It is straightforward from (27) that the (projected) ℓ^1 -penalized maximum likelihood estimate of θ is obtained by solving an optimization problem of the form (P) where ℓ and g are given by

$$\ell(\theta) = \frac{1}{N} \sum_{i=1}^N \langle \theta, \bar{B}(x^{(i)}) \rangle - \log Z_\theta, \quad \text{and} \quad g(\theta) = \lambda \sum_{1 \leq k < j \leq p} |\theta_{jk}| + \mathbb{I}_{\mathcal{K}_a}(\theta),$$

for a regularization parameter $\lambda > 0$, and where $\mathcal{K}_a \stackrel{\text{def}}{=} \{\theta \in \Theta : |\theta_{ij}| \leq a, \text{ for all } i, j\}$ for some constant $a > 0$. Notice that we do not penalize the diagonal terms of θ . By the Fisher identity, the gradient of the logarithm of the normalizing constant $\nabla \log Z_\theta$ is the expectation of the sufficient statistics $\bar{B}$, which implies

$$\nabla \ell(\theta) = \frac{1}{N} \sum_{i=1}^N \bar{B}(x^{(i)}) - \int_{\mathbb{X}^p} \bar{B}(z) \pi_\theta(dz), \quad (28)$$

where $\pi_\theta(dz) = f_\theta(z) \mu(dz)$, and f_θ is given by (27).

Proposition 11. *The function $\theta \mapsto \ell(\theta)$ is concave. and for all $\theta, \vartheta \in \Theta$,*

$$\|\nabla \ell(\theta) - \nabla \ell(\vartheta)\|_2 \leq \left(\sum_{j \leq i} \text{osc}^2(B_{ij}) \right) \|\theta - \vartheta\|_2, \quad (29)$$

where $\text{osc}(B_{ij}) = \sup_{x, y, u, v \in \mathbb{X}} |B(x, y) - B(u, v)|$ if $i \neq j$, and $\text{osc}(B_{ij}) = \sup_{x, y} |B_0(x) - B_0(y)|$, if $i = j$.

Proof. See Section 6.4. □

The proposition implies that H1 holds with $L = \sum_{j \leq i} \text{osc}^2(B_{ij})$. However an inspection of the proof of Proposition 11 shows that this value of L is very conservative. Although we do not pursue this here, a tighter bound can be derived by taking into account the sparsity of θ and ϑ . Since $\mathcal{K}_a$ is convex compact, and the log-likelihood function ℓ is concave continuous, and the ℓ^1 -penalty function is strictly convex and continuous, H2 holds as well and $\theta_\star$ is unique. Also, the representation of the gradient in (28) readily gives H3. Finally, since $\mathbb{X}$ is finite and $\mathcal{K}_a$ is compact, H4 holds as well.

The final ingredient needed to apply Algorithm 2 and Algorithm 3 is a method to generate random sample from π_θ . Unfortunately direct simulation from the Gibbs measures π_θ is rarely feasible, so we turn to MCMC. For $\theta \in \Theta$, we denote P_θ an ergodic Markov kernel on $\mathbf{X}^p$ with invariant distribution π_θ , and let ν_θ denote the initial distribution. These Markov kernels can be constructed in many ways. For instance the Gibbs sampler (see e.g. Robert and Casella (2005)) is a general algorithm that can be applied in many cases. However this algorithm often mixes poorly. Whenever possible we recommend the use of specialized algorithms with better mixing properties. We give an example in the next section. With the same notation as above, Algorithm 2 then becomes (Algorithm 3 translates similarly).

Algorithm 4 (Stochastic Proximal gradient algorithm for network estimation). At the n -th iteration, and given $\mathcal{F}_n = \sigma(\theta_1, \dots, \theta_n)$:

- (1) generate the $\mathbf{X}^p$ -valued Markov sequence $\{X_{n+1,j}\}_{j=0}^{m_{n+1}}$ with transition P_{θ_n} and initial distribution ν_{θ_n} , and compute the approximate gradient

$$H_{n+1} = \frac{1}{N} \sum_{i=1}^N \bar{B}(x^{(i)}) - \frac{1}{m_{n+1}} \sum_{j=1}^{m_{n+1}} \bar{B}(X_{n+1,j}),$$

- (2) Compute $\theta_{n+1} = \Pi_{\mathcal{K}_a}(s_{\gamma_{n+1}, \lambda, 1}(\theta_n + \gamma_{n+1} H_{n+1}))$, where as in Example 4, the operation $s_{\gamma, \lambda, 1}(M)$ soft-thresholds each entry of the matrix M , and the operation $\Pi_{\mathcal{K}_a}(M)$ projects each entry of M on $[-a, a]$.

5.1. Case of the Potts model. We focus on the particular case where $\mathbf{X} = \{1, \dots, M\}$, and $B(x, y) = \mathbb{1}_{\{x=y\}}$, which corresponds to the well known Potts model

$$f_\theta(x_1, \dots, x_p) = \frac{1}{Z_\theta} \exp \left\{ \sum_{i=1}^p \theta_{ii} B_0(x_i) + \sum_{1 \leq j < i \leq p} \theta_{ij} \mathbb{1}_{\{x_i = x_j\}} \right\}. \quad (30)$$

The term $\sum_{i=1}^p \theta_{ii} B_0(x_i)$ is sometimes referred to as the “external field” and defines the distribution in the absence of interaction. We focus on the case where the interactions terms θ_{ij} for $i \neq j$ are

nonnegative. This corresponds to networks with only collaborative interactions between the nodes.

We enforce this constraint by re-defining the set $\mathcal{K}_a$ as

$$\mathcal{K}_a = \{\theta \in \Theta : \theta_{ij} \in [-a, a], i \leq j, \text{ and } \theta_{ij} \geq 0, i \neq j\}.$$

With this specific example we now discuss how to design the Markov kernel P_θ , for a given $\theta \in \Theta$. For good mixing properties we advocate the use of the Wolff clustering algorithm (Wolff (1989)). The algorithm is a variant of the more well-known Swendsen-Wang algorithm which could also be used here; see Borgs et al. (2012) and the references therein. For $j \neq i$, we set $b_{ij} = e^{\theta_{ij}}$. Notice that $b_{ij} \geq 1$. Then for $x = (x_1, \dots, x_p)$, we can write

$$f_\theta(x) = \frac{1}{Z_\theta} \exp\left(\sum_{i=1}^p \theta_{ii} B_0(x_i)\right) \prod_{1 \leq j < i \leq p} \left(b_{ij} \mathbb{1}_{\{x_i = x_j\}} + \mathbb{1}_{\{x_i \neq x_j\}}\right).$$

Then we augment the variable x with binary auxiliary variables $\{\delta_{ij}, 1 \leq j < i \leq p\}$, $\delta_{ij} \in \{0, 1\}$; the joint distribution of (x, δ) is chosen to be

$$\begin{aligned} \bar{f}_\theta(x, \delta) \propto \exp\left(\sum_{i=1}^p \theta_{ii} B_0(x_i)\right) \\ \times \prod_{j < i} \left(\mathbb{1}_{\{x_i = x_j\}} b_{ij} \left(1 - b_{ij}^{-1}\right)^{\delta_{ij}} b_{ij}^{\delta_{ij}-1} + \mathbb{1}_{\{x_i \neq x_j\}} 0^{\delta_{ij}} 1^{1-\delta_{ij}}\right). \end{aligned}$$

It is clear from this expression that the marginal distribution of x in this joint distribution is the same f_θ given above. Note that the auxiliary variables $\{\delta_{ij}, 1 \leq j < i \leq p\}$ are conditionally independent given $x = (x_1, \dots, x_p)$ such that if $x_i \neq x_j$, then $\delta_{ij} = 0$ with probability 1, and if $x_i = x_j$, then $\delta_{ij} = 1$ with probability $1 - b_{ij}^{-1}$, and $\delta_{ij} = 0$ with probability b_{ij}^{-1} .

Now the variables $\{\delta_{ij}, 1 \leq j < i \leq p\}$ defines an undirected graph with nodes $\{1, \dots, p\}$ where there is an edge between $i \neq j$ if $\delta_{ij} = 1$, and there is no edge otherwise. This graph partitions the nodes $\{1, \dots, p\}$ into maximal clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$. By cluster, we mean either a single node, or a set of nodes where there is a path joining any two of them. A cluster is maximal if it loses the cluster property with the addition of any new node. Notice that $\delta_{ij} = 1$ implies $x_i = x_j$. Hence

all the nodes in a given cluster holds the same value of x . Therefore the conditional distribution $\bar{f}_\theta(x|\delta)$ writes

$$\bar{f}_\theta(x|\delta) \propto \prod_{k=1}^K \left[\exp \left(\sum_{i \in \mathcal{C}_k} \theta_{ii} B_0(x_i) \right) \prod_{j < i, (i,j) \in \mathcal{C}_k} \mathbb{1}_{\{x_i = x_j\}} \right].$$

Therefore given δ , the collections $(x_{\mathcal{C}_1}, \dots, x_{\mathcal{C}_K})$ are independent and can be simulated independently. The Wolff algorithm is the variant in which we randomly select a node $i \in \{1, \dots, p\}$ and grow the maximal cluster in which it belongs and update jointly all the nodes in that cluster. This leads to the following algorithm.

Algorithm 5 (Wolff algorithm to simulate from (30)). Given $X = (X_1, \dots, X_p)$

- (1) Randomly select $i \in \{1, \dots, p\}$, and set $\mathcal{C}_0 = \{i\}$.
- (2) Do until $\mathcal{C}_0$ can no longer grow. For each new addition j to $\mathcal{C}_0$, and for each $j' \notin \mathcal{C}_0$ such that $\theta_{jj'} > 0$, starting with $\delta_{jj'} = 0$, do the following. If $X_j = X_{j'}$, set $\delta_{jj'} = 1$ with probability $1 - e^{-\theta_{jj'}}$. If $\delta_{jj'} = 1$, add j' to $\mathcal{C}_0$.
- (3) If $X_i = v$, randomly select $v' \in \{1, \dots, M\} \setminus \{v\}$, and propose a new vector $\tilde{X} \in \mathbb{X}^p$, where $\tilde{X}_j = v'$ for $j \in \mathcal{C}_0$ and $\tilde{X}_j = X_j$ for $j \notin \mathcal{C}_0$. Accept $\tilde{X}$ with probability

$$1 \wedge \exp \left((B_0(v') - B_0(v)) \sum_{j \in \mathcal{C}_0} \theta_{jj} \right).$$

5.2. Simulation results. We report in this section some simulation results of the stochastic proximal gradient algorithm (Algorithm 4-Algorithm 5), and its accelerated version. We use $M = 20$, $B_0(x) = x$, $N = 500$ and for $p = 50, 100$, and 200 . We generate the “true” matrix $\theta_\star$ such that on average $s\%$ of the below diagonal are non-zeros where $s = 2$ for $p = 50$ and $s = 1$ for $p = 100$ and $p = 200$. All the diagonal elements are generated from $\mathbf{U}(-1, 1)$, and all the non-zero off-diagonal elements are simulated from $\mathbf{U}(4, 8)$. Notice that with $p = 200$, the model has 201×10^2 parameters and an average of 399 non-zeros parameters.

By trial-and-error we set the regularization parameter to $\lambda = 0.8\sqrt{\log(p)/n}$ for all the simulations. In a real data analysis where $\theta_\star$ is not available, the choice of λ typically incurs a significant

amount of computing, using cross-validation or a model selection criterion such as BIC. Given θ_* we simulate the dataset using the Wolff algorithm (Algorithm 5). For the Potts model the constant L in H1 as given by Proposition 11, can be taken as $L = (p(M-1)^2 + \frac{p^2-p}{2})$. However we found this value of L too conservative. In the simulations we use $L = p$ for the stochastic proximal gradient and $L = Mp$ for the Nesterov acceleration. We implement a fixed step-size algorithm with $\gamma_n = 1/L$. We use an increasing Monte Carlo batch size $m_n = 100 + n$ for the stochastic proximal gradient, and $m_n = 100 + n^{1.5}$ for the accelerated scheme. We recall from discussion in Section 4.2 that the accelerated scheme requires $m_n \propto n^\beta$ for $\beta > 1$ in order to converge.

For each simulation we report at each iteration of the algorithm the relative estimation error

$$\frac{\|\theta_n - \theta_*\|_2}{\|\theta_*\|_2},$$

the true positive rate (TPR) and the false discovery rate (FDR) of the off-diagonal elements of θ_n , defined as

$$\text{TPR}_n = \frac{\sum_{j < i} \mathbb{1}_{\{|\theta_{n,ij}| > 0\}} \mathbb{1}_{\{|\theta_{*,ij}| > 0\}}}{\sum_{j < i} \mathbb{1}_{\{|\theta_{*,ij}| > 0\}}}, \quad \text{FDR}_n = \frac{\sum_{j < i} \mathbb{1}_{\{|\theta_{n,ij}| > 0\}} \mathbb{1}_{\{\theta_{*,ij} = 0\}}}{\sum_{j < i} \mathbb{1}_{\{|\theta_{n,ij}| > 0\}}}.$$

The results are given on Figure 1–Figure 3. We first notice that the relative error remains high, even for $p = 50$. This confirm the fact that ℓ^1 -penalization methods, although very successful at structure/model selection, perform poorly in general for parameter estimation (due to the inherent severe bias). We notice also that the true positive rate is fairly good and converges very quickly. This suggests that the method detects very easily the non-zero off-diagonal components of θ_* . The false discovery rate decreases gradually to zero, but remains high for $p = 200$.

The comparison of the stochastic proximal algorithm and the accelerated algorithm is interesting. For $p = 50$, the accelerated scheme behaves well and vastly outperformed the plain algorithm (see Figure 1). However in general the accelerated scheme is unreliable and very sensitive to the choice of L in $\gamma_n = 1/L$. For instead for $p \in \{100, 200\}$ and $L = Mp$, the accelerated scheme initially appears to behave well, but eventually diverges (results not shown here). We also tested larger values of L

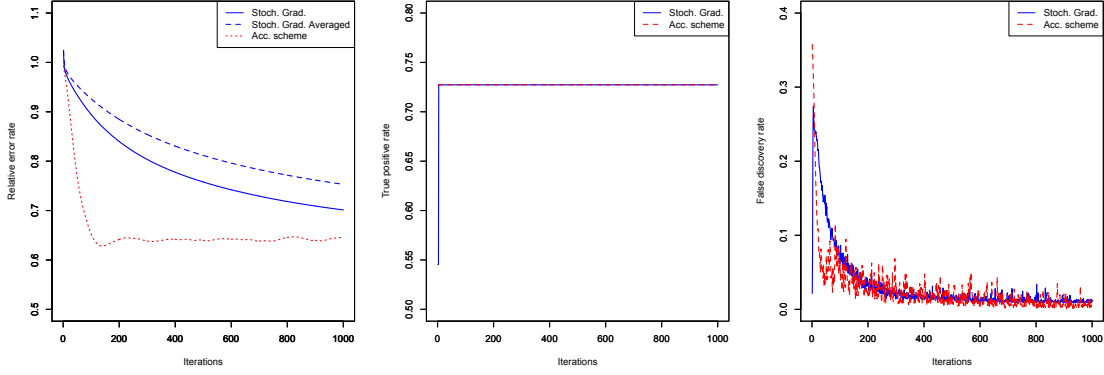


FIGURE 1. Stochastic proximal gradient and its accelerated version. $p = 50$.

without much success. For instance choosing L according to Proposition 11 ($L = p(M-1)^2 + \frac{p^2-p}{2}$) results in an accelerated scheme that converges but at a slower rate than the plain algorithm. These behaviors are consistent with Theorem 9 and Corollary 26 where we show that in general the accelerated scheme requires $m_n \sim n^{2+\epsilon}$ for some $\epsilon > 0$ to outperform the plain algorithm. However Monte Carlo batch-sizes of this magnitude are computationally very expensive for the current problem and were not tested.

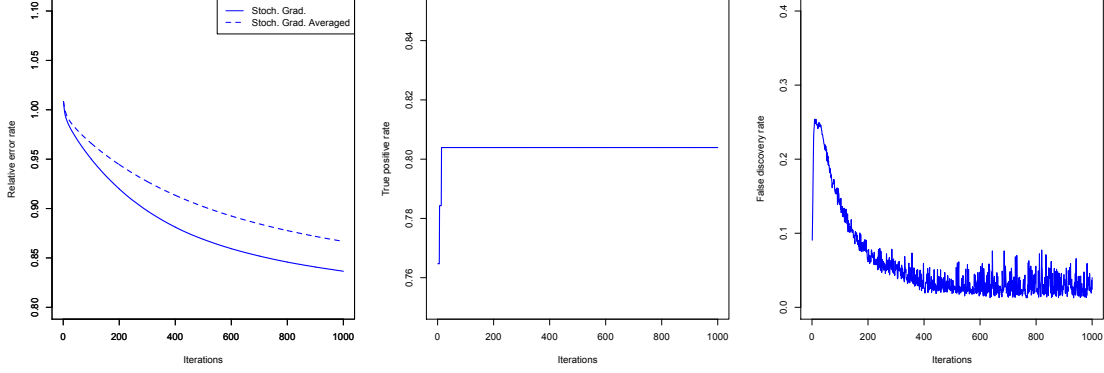
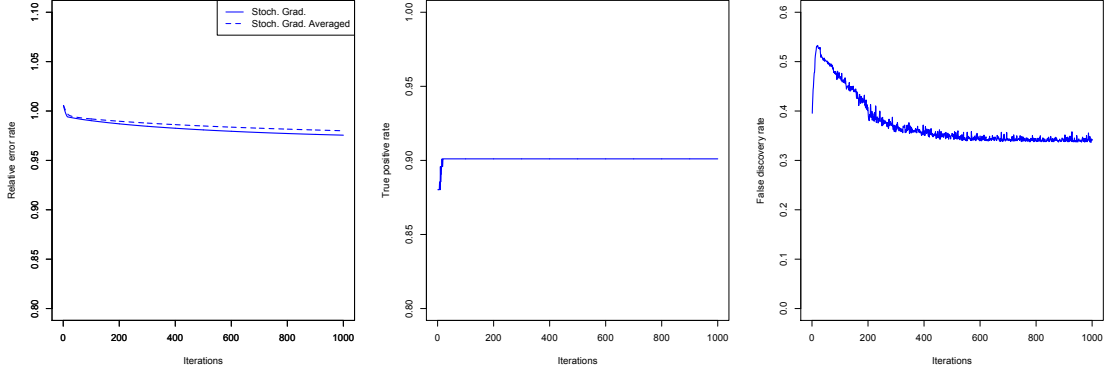
With the plain stochastic proximal gradient we also compare the uniform averaging estimate $\bar{\theta}_n$ and the non-averaged estimate θ_n . As shown on Figure 1–Figure 3 the non-averaged estimate performs better. By viewing the non-averaged estimate θ_n as a limiting case of the weighted averaging with $a_n = C_\alpha n^\alpha$ $\alpha > 0$ large, this result is consistent with our discussion in Section 3.2 where we show that when the gradient approximations are biased (as it is here), weighted averaging is preferable to the uniform averaging.

6. PROOFS

We will need the following properties of the proximal operator.

Lemma 12. *For $\theta \in \Theta$, $\gamma > 0$, and $\vartheta \in \Theta$,*

$$\gamma \{g(\text{Prox}_\gamma(\theta)) - g(\vartheta)\} \leq \langle \vartheta - \text{Prox}_\gamma(\theta), \text{Prox}_\gamma(\theta) - \theta \rangle . \quad (31)$$

FIGURE 2. Stochastic proximal gradient. $p = 100$.FIGURE 3. Stochastic proximal gradient. $p = 200$.

For any $\gamma > 0$, the operator $\theta \mapsto \text{Prox}_\gamma(\theta)$ is firmly nonexpansive, i.e. for any $\theta, \vartheta \in \Theta$,

$$\langle \text{Prox}_\gamma(\theta) - \text{Prox}_\gamma(\vartheta), \theta - \vartheta \rangle \geq \|\text{Prox}_\gamma(\theta) - \text{Prox}_\gamma(\vartheta)\|^2. \quad (32)$$

In particular, the maps $\theta \mapsto \text{Prox}_\gamma(\theta)$ and $\theta \mapsto \theta - \text{Prox}_\gamma(\theta)$ are Lipschitz with Lipschitz constants that can be taken equal to 1.

Proof. See (Bauschke and Combettes, 2011, Propositions 12.26 and 12.27). $\square$

Lemma 12 has the following useful consequence. Set

$$F(\theta) \stackrel{\text{def}}{=} -\ell(\theta) + g(\theta). \quad (33)$$

Lemma 13. *Assume H1 and let $\gamma \in (0, 1/L]$. Then for all $\theta, \vartheta \in \Theta$,*

$$F(\text{Prox}_\gamma(\theta)) - F(\vartheta) \leq -\frac{1}{2\gamma} \|\text{Prox}_\gamma(\theta) - \vartheta\|^2 + \frac{1}{\gamma} \langle \vartheta - \text{Prox}_\gamma(\theta), \vartheta + \gamma \nabla \ell(\vartheta) - \theta \rangle, \quad (34)$$

where F is defined in (33). For all $\theta, \vartheta, \xi \in \Theta$,

$$\begin{aligned} F(\text{Prox}_\gamma(\xi)) - F(\vartheta) &\leq \frac{1}{2\gamma} (\|\vartheta - \theta\|^2 - \|\vartheta - \text{Prox}_\gamma(\xi)\|^2 \\ &\quad + 2 \langle \vartheta - \text{Prox}_\gamma(\xi), \theta + \gamma \nabla \ell(\theta) - \xi \rangle). \end{aligned} \quad (35)$$

Proof. Let $\theta, \vartheta \in \Theta$. Set $q(\theta|\vartheta) \stackrel{\text{def}}{=} -\ell(\vartheta) - \langle \nabla \ell(\vartheta), \theta - \vartheta \rangle + \frac{1}{2\gamma} \|\theta - \vartheta\|^2$. Notice that under H1, when $\gamma^{-1} \geq L$, then $-\ell(\theta) \leq q(\theta|\vartheta)$ for all $\theta, \vartheta \in \Theta$. Therefore, by applying this inequality with $\theta \leftarrow \text{Prox}_\gamma(\theta)$, we have

$$-\ell(\text{Prox}_\gamma(\theta)) \leq -\ell(\vartheta) - \langle \nabla \ell(\vartheta), \text{Prox}_\gamma(\theta) - \vartheta \rangle + \frac{1}{2\gamma} \|\text{Prox}_\gamma(\theta) - \vartheta\|^2. \quad (36)$$

Combine this with (31) to get

$$\begin{aligned} F(\text{Prox}_\gamma(\theta)) - F(\vartheta) &= -\ell(\text{Prox}_\gamma(\theta)) + g(\text{Prox}_\gamma(\theta)) - (-\ell(\vartheta) + g(\vartheta)) \\ &\leq \frac{1}{2\gamma} \|\text{Prox}_\gamma(\theta) - \vartheta\|^2 + \frac{1}{\gamma} \langle \vartheta - \text{Prox}_\gamma(\theta), \gamma \nabla \ell(\vartheta) + \text{Prox}_\gamma(\theta) - \theta \rangle \\ &= -\frac{1}{2\gamma} \|\text{Prox}_\gamma(\theta) - \vartheta\|^2 + \frac{1}{\gamma} \langle \vartheta - \text{Prox}_\gamma(\theta), \vartheta + \gamma \nabla \ell(\vartheta) - \theta \rangle. \end{aligned}$$

This concludes the proof of (34). By (36) again,

$$\begin{aligned} &-\ell(\text{Prox}_\gamma(\xi)) + \ell(\vartheta) \\ &\leq -\ell(\theta) - \langle \nabla \ell(\theta), \text{Prox}_\gamma(\xi) - \theta \rangle + \frac{1}{2\gamma} \|\text{Prox}_\gamma(\xi) - \theta\|^2 + \ell(\vartheta) \\ &= [-\ell(\theta) - \langle \nabla \ell(\theta), \vartheta - \theta \rangle + \ell(\vartheta)] - \langle \nabla \ell(\theta), \text{Prox}_\gamma(\xi) - \vartheta \rangle + \frac{1}{2\gamma} \|\text{Prox}_\gamma(\xi) - \theta\|^2 \\ &\leq -\langle \nabla \ell(\theta), \text{Prox}_\gamma(\xi) - \vartheta \rangle + \frac{1}{2\gamma} \|\text{Prox}_\gamma(\xi) - \theta\|^2, \end{aligned}$$

using the convexity of $-\ell$. Combine this inequality with (31) to get

$$\begin{aligned}
& F(\text{Prox}_\gamma(\xi)) - F(\vartheta) \\
& \leq \frac{1}{2\gamma} \|\text{Prox}_\gamma(\xi) - \theta\|^2 + \frac{1}{\gamma} \langle \vartheta - \text{Prox}_\gamma(\xi), \text{Prox}_\gamma(\xi) - \xi + \gamma \nabla \ell(\theta) \rangle \\
& = \frac{1}{2\gamma} (\|\vartheta - \theta\|^2 - \|\vartheta - \text{Prox}_\gamma(\xi)\|^2 + 2 \langle \vartheta - \text{Prox}_\gamma(\xi), \theta - \xi + \gamma \nabla \ell(\theta) \rangle) .
\end{aligned}$$

□

For $\theta \in \Theta$, $\gamma > 0$, set

$$T_\gamma(\theta) \stackrel{\text{def}}{=} \text{Prox}_\gamma(\theta + \gamma \nabla \ell(\theta)) , \quad (37)$$

the gradient proximal map.

Lemma 14. *Assume H2. Then for any $\gamma \in (0, 2/L]$,*

$$\|\theta + \gamma \nabla \ell(\theta) - \vartheta - \gamma \nabla \ell(\vartheta)\| \leq \|\theta - \vartheta\| , \quad (38)$$

$$\|T_\gamma(\theta) - T_\gamma(\vartheta)\| \leq \|\theta - \vartheta\| . \quad (39)$$

Proof. Since ℓ is a concave function with Lipschitz-continuous gradients, (Nesterov, 2004, Theorem 2.1.5) shows that, for all $\theta, \vartheta \in \Theta$,

$$\langle \nabla \ell(\theta) - \nabla \ell(\vartheta), \theta - \vartheta \rangle \leq -\frac{1}{L} \|\nabla \ell(\theta) - \nabla \ell(\vartheta)\|^2 .$$

it holds for any $\theta, \vartheta \in \Theta$

$$\begin{aligned}
& \|\theta + \gamma \nabla \ell(\theta) - \vartheta - \gamma \nabla \ell(\vartheta)\|^2 \\
& \leq \|\theta - \vartheta\|^2 + \gamma^2 \|\nabla \ell(\theta) - \nabla \ell(\vartheta)\|^2 + 2\gamma \langle \nabla \ell(\theta) - \nabla \ell(\vartheta), \theta - \vartheta \rangle \\
& \leq \|\theta - \vartheta\|^2 + \gamma \left(\gamma - \frac{2}{L} \right) \|\nabla \ell(\theta) - \nabla \ell(\vartheta)\|^2 .
\end{aligned}$$

Since $\gamma - 2/L \leq 0$, (38) follows. The proof of (39) follows from the Lipschitz property of the proximal map Prox_γ (see Lemma 12) and (38). □

Lemma 15. *Assume H1 and let $\gamma \in (0, 1/L]$. For $\theta \in \Theta$ and any $H \in \mathbb{R}^d$, we get*

$$|F[\text{Prox}_\gamma(\theta + \gamma H)] - F[T_\gamma(\theta)]| \leq \|\eta\| (\gamma\|\eta\| + \|\theta - T_\gamma(\theta)\|) , \quad (40)$$

where $\eta \stackrel{\text{def}}{=} H - \nabla\ell(\theta)$ and F and T_γ are defined in (33) and (37), respectively.

Proof. Set $S_\gamma(\theta) \stackrel{\text{def}}{=} \text{Prox}_\gamma(\theta + \gamma H)$. Apply (34) with $\vartheta \leftarrow T_\gamma(\theta)$ and $\theta \leftarrow \theta + \gamma H$:

$$F(S_\gamma(\theta)) - F(T_\gamma(\theta)) \quad (41)$$

$$\begin{aligned} &\leq -\frac{1}{2\gamma} \|S_\gamma(\theta) - T_\gamma(\theta)\|^2 + \frac{1}{\gamma} \langle T_\gamma(\theta) - S_\gamma(\theta), T_\gamma(\theta) + \gamma \nabla\ell(T_\gamma(\theta)) - \theta - \gamma H \rangle \\ &\leq \frac{1}{\gamma} \|T_\gamma(\theta) - S_\gamma(\theta)\| (\|T_\gamma(\theta) + \gamma \nabla\ell(T_\gamma(\theta)) - \theta - \gamma \nabla\ell(\theta)\| + \gamma\|\eta\|) . \end{aligned} \quad (42)$$

Since $\theta \mapsto \text{Prox}_\gamma(\theta)$ is a Lipschitz contraction (see Lemma 12), we have

$$\|T_\gamma(\theta) - S_\gamma(\theta)\| = \|\text{Prox}_\gamma(\theta + \gamma \nabla\ell(\theta)) - \text{Prox}_\gamma(\theta + \gamma H)\| \leq \gamma\|\eta\| . \quad (43)$$

By plugging (43) and (38) in (42), we get

$$F(S_\gamma(\theta)) - F(T_\gamma(\theta)) \leq \|\eta\| (\gamma\|\eta\| + \|\theta - T_\gamma(\theta)\|) .$$

Similarly, we apply the same inequality with $\vartheta \leftarrow S_\gamma(\theta)$, and $\theta \leftarrow \theta + \gamma \nabla\ell(\theta)$ to get

$$\begin{aligned} F(S_\gamma(\theta)) - F(T_\gamma(\theta)) &\geq \frac{1}{\gamma} \langle T_\gamma(\theta) - S_\gamma(\theta), S_\gamma(\theta) + \gamma \nabla\ell(S_\gamma(\theta)) - \theta - \gamma \nabla\ell(\theta) \rangle \\ &\geq -\|\eta\| \|\theta - S_\gamma(\theta)\| \\ &\geq -\|\eta\| (\|\theta - T_\gamma(\theta)\| + \gamma\|\eta\|) , \end{aligned}$$

where in the last inequality, we use again (43) and (38). The results follows by combining the two bounds. $\square$

It is an easy consequence of Lemma 13 that $\theta \mapsto F(\theta)$ (resp. under H2, the function $\theta \mapsto V(\theta)$ defined by (8)) is a Lyapunov function (resp. a non-negative Lyapunov function) for the gradient proximal map T_γ given by (37).

Lemma 16. *Assume H1 and H2. Suppose that $\gamma \in (0, 1/L]$. Then*

$$V(T_\gamma(\theta)) \leq V(\theta) - \frac{1}{2\gamma} \|\theta - T_\gamma(\theta)\|^2, \quad \theta \in \Theta. \quad (44)$$

Proof. Apply (35) with $\vartheta \leftarrow \theta$, and $\xi \leftarrow \theta + \gamma \nabla \ell(\theta)$. $\square$

6.1. Proof of Theorem 5.

Proof. Under H1, V is convex so that

$$V(\bar{\theta}_n) \leq \left(\sum_{k=1}^n a_k \right)^{-1} \sum_{k=1}^n a_k V(\theta_k). \quad (45)$$

We first apply (35) with $\xi \leftarrow \theta + \gamma_{j+1} H_{j+1}$ and $\theta \leftarrow \theta_j$, $\vartheta \leftarrow \theta_\star$, $\gamma \leftarrow \gamma_{j+1}$:

$$V(\theta_{j+1}) - V(\theta_\star) \leq \frac{1}{2\gamma_{j+1}} (\|\theta_j - \theta_\star\|^2 - \|\theta_{j+1} - \theta_\star\|^2) + \langle \theta_{j+1} - \theta_\star, \eta_{j+1} \rangle.$$

Noting that $V(\theta_\star) = 0$, and multiplying both side by a_{j+1} gives:

$$\begin{aligned} a_{j+1} V(\theta_{j+1}) &\leq \frac{1}{2} \left(\frac{a_{j+1}}{\gamma_{j+1}} - \frac{a_j}{\gamma_j} \right) \|\theta_j - \theta_\star\|^2 + \frac{a_j}{2\gamma_j} \|\theta_j - \theta_\star\|^2 \\ &\quad - \frac{a_{j+1}}{2\gamma_{j+1}} \|\theta_{j+1} - \theta_\star\|^2 + a_{j+1} \langle \theta_{j+1} - \theta_\star, \eta_{j+1} \rangle. \end{aligned}$$

Summing from $j = 0$ to $n - 1$, with $a_0 = 0$, and taking the expectation gives

$$\sum_{j=1}^n a_j \mathbb{E}[V(\theta_j)] \leq \frac{1}{2} \sum_{j=1}^n \left(\frac{a_j}{\gamma_j} - \frac{a_{j-1}}{\gamma_{j-1}} \right) \mathbb{E}[\|\theta_{j-1} - \theta_\star\|^2] + \sum_{j=1}^n a_j \mathbb{E}[\langle \theta_j - \theta_\star, \eta_j \rangle] \quad (46)$$

Let T_γ be given by (37). We decompose $\langle \theta_j - \theta_\star, \eta_j \rangle$ as follows:

$$\langle \theta_j - \theta_\star, \eta_j \rangle = \langle \theta_j - T_{\gamma_j}(\theta_{j-1}), \eta_j \rangle + \langle T_{\gamma_j}(\theta_{j-1}) - \theta_\star, \eta_j \rangle.$$

Since the proximal map is 1-Lipschitz (see Lemma 12), we get

$$|\langle \theta_j - T_{\gamma_j}(\theta_{j-1}), \eta_j \rangle| \leq \gamma_j \|\eta_j\|^2.$$

By conditioning on $\mathcal{F}_{j-1}$ and using that $T_{\gamma_j}(\theta_\star) = \theta_\star$ and T_{γ_j} is a 1-Lipschitz operator (see Lemma 14), we have

$$\begin{aligned} |\mathbb{E} [\langle T_{\gamma_j}(\theta_{j-1}) - \theta_\star, \eta_j \rangle \mid \mathcal{F}_{j-1}]| &= |\langle T_{\gamma_j}(\theta_{j-1}) - T_{\gamma_j}(\theta_\star), \mathbb{E} [\eta_j \mid \mathcal{F}_{j-1}] \rangle| \\ &\leq \|\theta_{j-1} - \theta_\star\| \|\mathbb{E} [\eta_j \mid \mathcal{F}_{j-1}]\| = \|\theta_{j-1} - \theta_\star\| \epsilon_{j-1}^{(1)}. \end{aligned}$$

Consequently,

$$|\mathbb{E} [\langle \theta_j - \theta_\star, \eta_j \rangle]| \leq \mathbb{E} [\|\theta_{j-1} - \theta_\star\| \epsilon_{j-1}^{(1)}] + \gamma_j \mathbb{E} [\epsilon_{j-1}^{(2)}].$$

This latter bound together with (45) and (46) yield the stated result. $\square$

6.2. Proof Theorem 8. The main ingredient is the deterministic result for random iterative maps developed in (Fort and Moulines, 2003, Proposition 9). Indeed, Lemma 16 shows that V is a Lyapunov function for the map T_γ with respect to $\mathcal{L}$: for any $\theta \in \Theta$, $V(T_\gamma(\theta)) - V(\theta) \leq 0$ and for any compact $\mathcal{K} \subseteq \Theta \setminus \mathcal{L}$, $\inf_{\mathcal{K}} V \circ T_\gamma - V < 0$.

Assume that (the proof is postponed below) for any $M > 0$,

$$\lim_{n \rightarrow \infty} |V(\theta_{n+1}) - V(T_\gamma(\theta_n))| \mathbb{1}_{\|\theta_n - \theta_\star\| \leq M} = 0, \quad \mathbb{P} - a.s. \quad (47)$$

Since $\{\theta_n, n \geq 0\}$ remains almost surely in a compact set by assumption, the theorem follows from (Fort and Moulines, 2003, Proposition 9). We now prove (47). Using Lemma 15 we get

$$|V(\theta_{n+1}) - V(T_\gamma(\theta_n))| \leq \gamma \|\eta_{n+1}\|^2 + \|\eta_{n+1}\| \|\theta_n - T_\gamma(\theta_n)\|.$$

Under H2, $\theta_n - T_\gamma(\theta_n) = \theta_n - \theta_\star - T_\gamma(\theta_n) + T_\gamma(\theta_\star)$, and since the T_γ is a Lipschitz function, we conclude that there exists $c > 0$ such that for any $n \geq 1$, almost-surely,

$$|V(\theta_{n+1}) - V(T_\gamma(\theta_n))| \leq \gamma \|\eta_{n+1}\|^2 + c \|\eta_{n+1}\|.$$

Hence it suffices to show that $\lim_n \|\eta_{n+1}\| \mathbb{1}_{\|\theta_n - \theta_\star\| \leq M} = 0$ almost surely. It suffices to show that for all $\delta > 0$, $\lim_{n \rightarrow \infty} \mathbb{P}[\sup_{j \geq n} \|\eta_{j+1}\| \mathbb{1}_{\|\theta_j - \theta_\star\| \leq M} > \delta] = 0$. We have

$$\begin{aligned} \mathbb{P} \left[\sup_{j \geq n} \|\eta_{j+1}\| \mathbb{1}_{\|\theta_j - \theta_\star\| \leq M} > \delta \right] &\leq \sum_{j \geq n} \mathbb{P} \left[\|\eta_{j+1}\| \mathbb{1}_{\{\|\theta_j - \theta_\star\| \leq M\}} > \delta \right] \\ &\leq \frac{M^p}{\delta^p} \sum_{j \geq n} \mathbb{E} \left[\|\eta_{j+1}\|^p \mathbb{1}_{\{\|\theta_j - \theta_\star\| \leq M\}} \right]. \end{aligned}$$

Taking the limit as $n \rightarrow \infty$ and using H4, it follows that

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\sup_{j \geq n} \|\eta_{j+1}\| \mathbb{1}_{\|\theta_j - \theta_\star\| \leq M} > \delta \right] = 0,$$

which concludes the proof.

6.3. Proof of Theorem 9. For $j \geq 0$, set $u_j = (1 - t_j^{-1})\theta_j + t_j^{-1}\theta_\star$ and $\eta_{j+1} = H_{j+1} - \nabla \ell(\vartheta_j)$ where ϑ_j is given in (22). Apply (35) with $\vartheta \leftarrow u_j$, $\theta \leftarrow \vartheta_j$, $\xi \leftarrow \vartheta_j + \gamma_{j+1}H_{j+1}$ and $\gamma \leftarrow \gamma_{j+1}$, to get

$$V(\theta_{j+1}) - V(u_j) \leq \frac{1}{2\gamma_{j+1}} (\|u_j - \vartheta_j\|^2 - \|u_j - \theta_{j+1}\|^2) + \frac{1}{\gamma_{j+1}} \langle u_j - \theta_{j+1}, -\gamma_{j+1}\eta_{j+1} \rangle.$$

By convexity of V , and since $V(\theta_\star) = 0$, $-V((1 - t_j^{-1})\theta_j + t_j^{-1}\theta_\star) \geq -(1 - t_j^{-1})V(\theta_j)$, so that

$$V(\theta_{j+1}) - (1 - t_j^{-1})V(\theta_j) \leq \frac{1}{2\gamma_{j+1}} (\|u_j - \vartheta_j\|^2 - \|u_j - \theta_{j+1}\|^2) - \langle u_j - \theta_{j+1}, \eta_{j+1} \rangle.$$

We check that $u_j - \vartheta_j = t_j^{-1}(-\theta_{j-1} - t_{j-1}(\theta_j - \theta_{j-1}) + \theta_\star) = -t_j^{-1}\Delta_{j-1}$, where $\Delta_j \stackrel{\text{def}}{=} \theta_j + t_j(\theta_{j+1} - \theta_j) - \theta_\star$. We also check that $u_j - \theta_{j+1} = -t_j^{-1}\Delta_j$. We conclude that

$$V(\theta_{j+1}) - (1 - t_j^{-1})V(\theta_j) \leq \frac{1}{2\gamma_{j+1}t_j^2} (\|\Delta_{j-1}\|^2 - \|\Delta_j\|^2) + \frac{1}{t_j} \langle \Delta_j, \eta_{j+1} \rangle.$$

Multiply both side by $\gamma_{j+1}t_j^2$ and noticing that $\gamma_{j+1}t_j^2(1 - t_j^{-1}) \leq \gamma_j t_{j-1}^2$ (by (20) and the assumptions on the sequence $\{\gamma_n, n \geq 1\}$), we get

$$\gamma_{j+1}t_j^2V(\theta_{j+1}) - \gamma_j t_{j-1}^2V(\theta_j) \leq \frac{1}{2} (\|\Delta_{j-1}\|^2 - \|\Delta_j\|^2) + \gamma_{j+1}t_j \langle \Delta_j, \eta_{j+1} \rangle.$$

Summing up for $j = 1$ to n and computing the expectation (remember that $t_0 = 1$)

$$\gamma_{n+1} t_n^2 \mathbb{E}[V(\theta_{n+1})] \leq \mathbb{E} \left[\gamma_1 V(\theta_1) + \frac{1}{2} \|\theta_1 - \theta_\star\|^2 \right] + \sum_{j=1}^n \gamma_{j+1} t_j \mathbb{E}[\langle \Delta_j, \eta_{j+1} \rangle] .$$

From the definition of Δ_j we have:

$$\begin{aligned} \langle \Delta_j, \eta_{j+1} \rangle &= \langle \theta_j - \theta_\star, \eta_{j+1} \rangle + t_j \langle \theta_{j+1} - \theta_j, \eta_{j+1} \rangle \\ &= (1 - t_j) \langle \theta_j - \theta_\star, \eta_{j+1} \rangle + t_j \langle \theta_{j+1} - T_{\gamma_{j+1}}(\vartheta_j), \eta_{j+1} \rangle \\ &\quad + t_j \langle T_{\gamma_{j+1}}(\vartheta_j) - \theta_\star, \eta_{j+1} \rangle . \end{aligned}$$

Since $|1 - t_j| = t_j - 1 \leq t_j$ and the proximal map (resp. the gradient proximal map) is 1-Lipschitz (see Lemma 12 and Lemma 14)

$$|\mathbb{E}[\langle \Delta_j, \eta_{j+1} \rangle]| \leq t_j \mathbb{E}[\|\theta_j - \theta_\star\| \epsilon_j^{(1)}] + \gamma_{j+1} t_j \mathbb{E}[\epsilon_j^{(2)}] + t_j \mathbb{E}[\|\vartheta_j - \theta_\star\| \epsilon_j^{(1)}] .$$

The theorem follows.

6.4. Proof of Proposition 11. To prove that $\theta \mapsto \ell(\theta)$ is concave, it suffices to show that $\theta \mapsto \log Z(\theta)$ is convex (since $\theta \mapsto n^{-1} \sum_{i=1}^n \langle \theta, \bar{B}(X^{(i)}) \rangle$ is linear). This is easy since for any $\gamma \in [0, 1]$,

$$\begin{aligned} &\gamma \log Z(\theta) + (1 - \gamma) \log Z(\vartheta) \\ &= \log \left[Z(\vartheta) \left(\int_{\mathbf{X}^p} \exp(\langle \bar{B}(x), \theta \rangle - \langle \bar{B}(x), \vartheta \rangle) \frac{e^{\langle \bar{B}(x), \vartheta \rangle}}{Z(\vartheta)} \mu(dx) \right)^\gamma \right] \\ &\geq \log \left[\int_{\mathbf{X}^p} \exp(\gamma \langle \bar{B}(x), \theta \rangle + (1 - \gamma) \langle \bar{B}(x), \vartheta \rangle) \mu(dx) \right] = \log Z(\gamma\theta + (1 - \gamma)\vartheta) . \end{aligned}$$

For $\theta, \vartheta \in \Theta$, the (i, j) -th entry of the matrix $\nabla \ell(\theta) - \nabla \ell(\vartheta)$ is given by

$$(\nabla \ell(\theta) - \nabla \ell(\vartheta))_{ij} = \int_{\mathbf{X}^p} B_{ij}(x_i, x_j) \pi_\vartheta(dx) - \int_{\mathbf{X}^p} B_{ij}(x_i, x_j) \pi_\theta(dx) .$$

For $t \in [0, 1]$ let

$$\pi_t(dz) \stackrel{\text{def}}{=} \exp(\langle \bar{B}(z), t\vartheta + (1-t)\theta \rangle) / \int \exp(\langle \bar{B}(x), t\vartheta + (1-t)\theta \rangle) \mu(dx),$$

defines a probability measure on $\mathbf{X}^p$. It is straightforward to check that

$$(\nabla \ell(\theta) - \nabla \ell(\vartheta))_{ij} = \int B_{ij}(x_i, x_j) \pi_1(dx) - \int B_{ij}(x_i, x_j) \pi_0(dx),$$

and that $t \mapsto \int B_{ij}(x_i, x_j) \pi_t(dx)$ is differentiable with derivative

$$\begin{aligned} \frac{d}{dt} \int B_{ij}(x_i, x_j) \pi_t(dx) &= \int B_{ij}(x_i, x_j) \left\langle \bar{B}(x) - \int \bar{B}(z) \pi_t(dz), \vartheta - \theta \right\rangle \pi_t(dx), \\ &= \text{Cov}_t(B_{ij}(X_i, X_j), \langle \bar{B}(X), \vartheta - \theta \rangle), \end{aligned}$$

where the covariance is taken assuming that $X \sim \pi_t$. Hence

$$\begin{aligned} |(\nabla \ell(\theta) - \nabla \ell(\vartheta))_{ij}| &= \left| \int_0^1 dt \text{Cov}_t(B_{ij}(X_i, X_j), \langle \bar{B}(X), \vartheta - \theta \rangle) \right| \\ &\leq \text{osc}(B_{ij}) \sqrt{\sum_{k \leq l} \text{osc}^2(B_{kl})} \|\theta - \vartheta\|_2. \end{aligned}$$

This implies the stated result.

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